Design and Verification of a Parameterized UART Transmitter Core

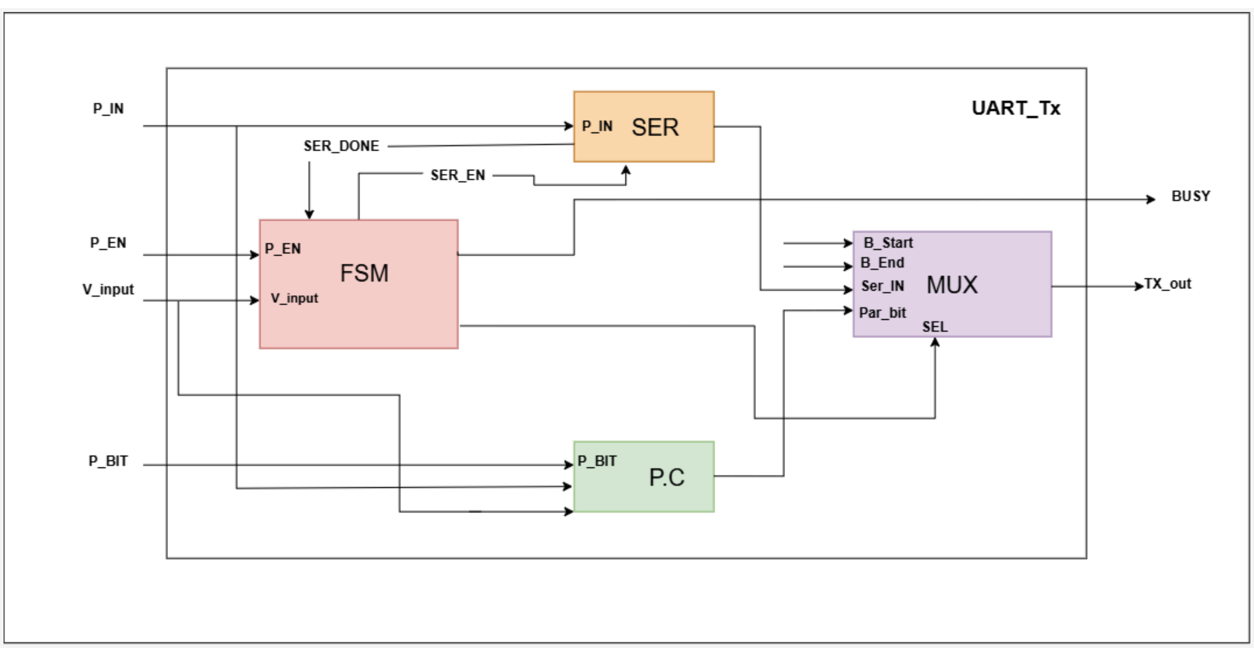
Abstract

This report details the design, implementation, and unit verification of a fully parameterized Universal Asynchronous Receiver-Transmitter (UART) Transmitter engine implemented in Verilog HDL. The architecture consists of four distinct sub-modules: a Finite State Machine (FSM) Controller, a Parallel-to-Serial Converter (Serializer), a Parity Calculator (P.C), and a 4-to-1 Multiplexer (MUX). Each constituent module was verified independently using a dedicated testbench before full top-level integration.

1. System Architecture Overview

The top-level UART_TX architecture transforms parallel data (P_IN) into a serial bitstream (TX_out) framed by standard Start (0), Parity (Par_bit), and Stop (1) bits.

Figure 1: Top-Level Architectural Block Diagram



Module Descriptions

Module Name	Identifier	Description
FSM Controller	FSM_control	Coordinates framing transitions and asserts system control signals (Ser_En, Sel, Busy) based on inputs P_EN, V_input, and Ser_done.
Serializer	Serializer	Performs parallel-to-serial conversion of data byte P_IN when enabled by Ser_En, generating Ser_done upon completion.
Parity Calculator	Par_calc	Calculates even or odd parity over the data byte P_IN as selected by configuration bit P_Bit.
Multiplexer	MUX	Routes the appropriate line—Start Bit (0), Serial Bit (Ser_IN), Parity Bit (Par_bit), or Stop Bit (1)—to output port TX_out based on selector code Sel.

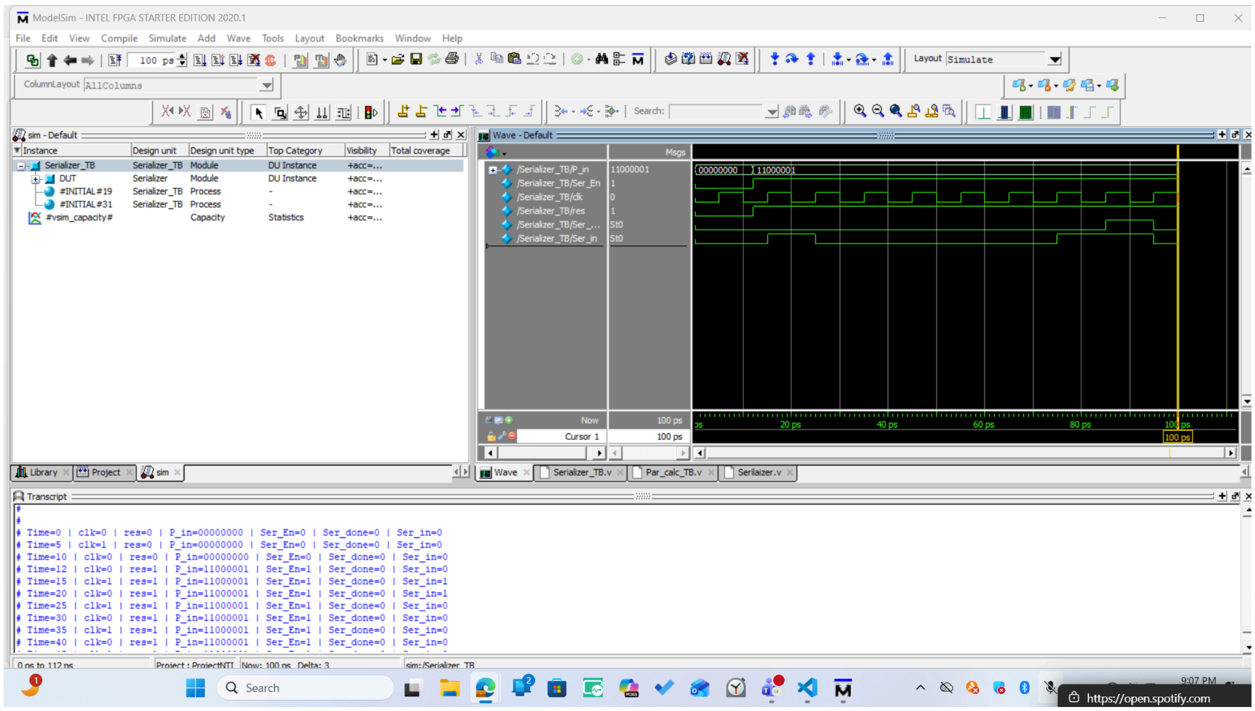
2. Unit Verification and Simulation Waveform Analysis

2.1 Serializer Sub-Module Verification

Figure 2: Serializer Module Waveform

The Serializer module converts parallel input data P_IN to a single-bit serial stream Ser_IN.

- **Execution Flow:** Upon asserting enable signal Ser_En = 1, the internal shift register loads P_IN = 8'b11000001. On subsequent clock cycles, bits are shifted out LSB-first over Ser_IN.
- **Waveform Commentary:** Ser_IN sequentially reflects bits 1 -> 0 -> 0 -> 0 -> 0 -> 0 -> 1 -> 1. Flag Ser_done pulses high upon completing the transfer sequence.



2.2 Multiplexer (MUX) Sub-Module Verification

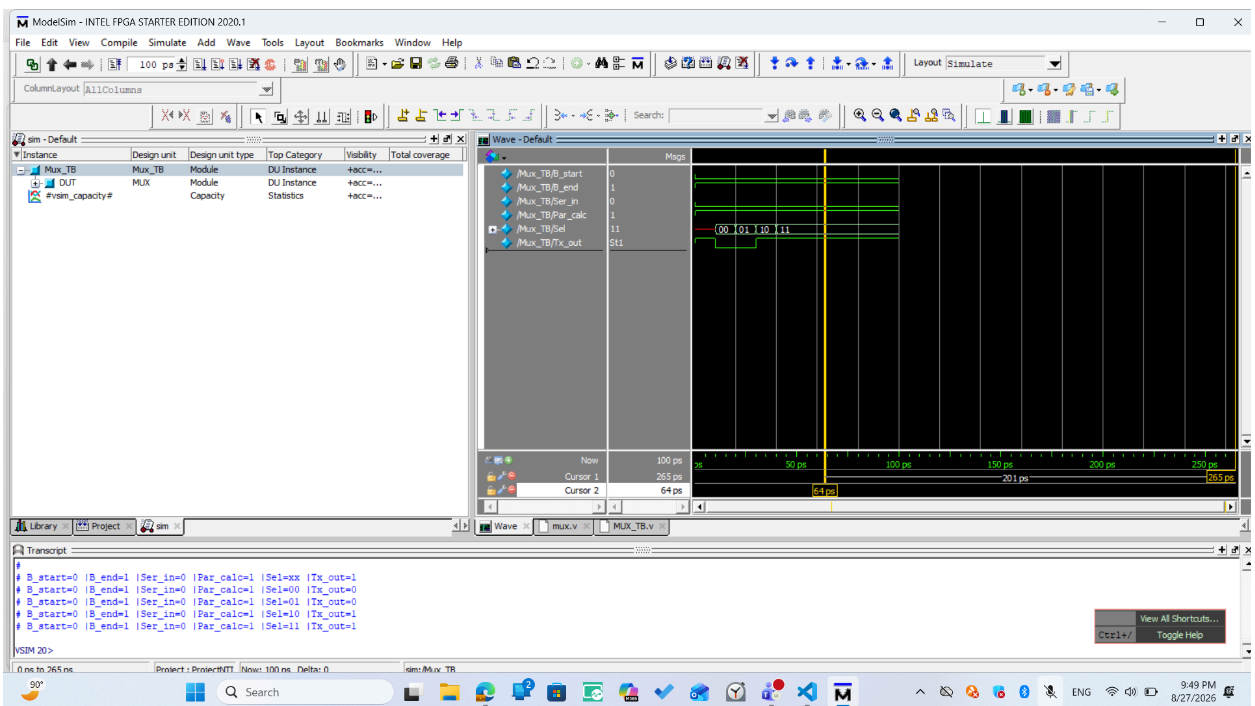
Figure 3: Multiplexer Waveform

The MUX module selects output line TX_out based on selector signal Sel[1:0].

Selection Mapping:

Selector Signal (Sel[1:0])	Routed Line	Output (Tx_out)
2'b00	Start Bit	0
2'b01	Stop Bit	1
2'b10	Calculated Parity Bit	Par_calc
2'b11	Serial Bit Stream	Ser_IN

Waveform Commentary: As Sel transitions from 00 through 11, output Tx_out mirrors the corresponding input line accurately without logic glitches.

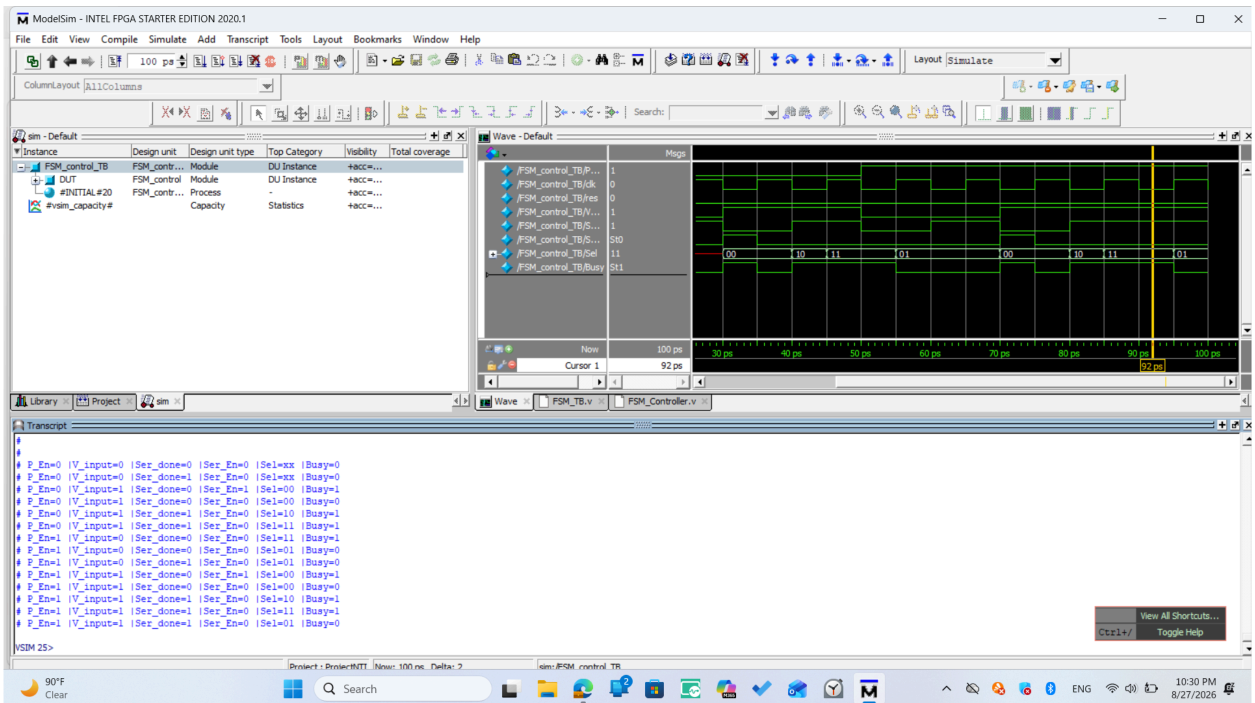


2.3 FSM Controller Sub-Module Verification

Figure 4: FSM Controller Waveform

The FSM controller governs state transitions across IDLE (00), DATA (11), PARITY (10), and STOP (01).

- **Execution Flow:** When valid transaction trigger V_input = 1 occurs in IDLE, state register transitions to DATA (Sel = 11) and asserts status line Busy = 1. Upon receiving Ser_done = 1, state advances to PARITY (Sel = 10), then STOP (Sel = 01), and back to IDLE (Busy = 0).
- **Waveform Commentary:** Signal Busy remains asserted throughout frame construction and de-asserts upon returning to IDLE.

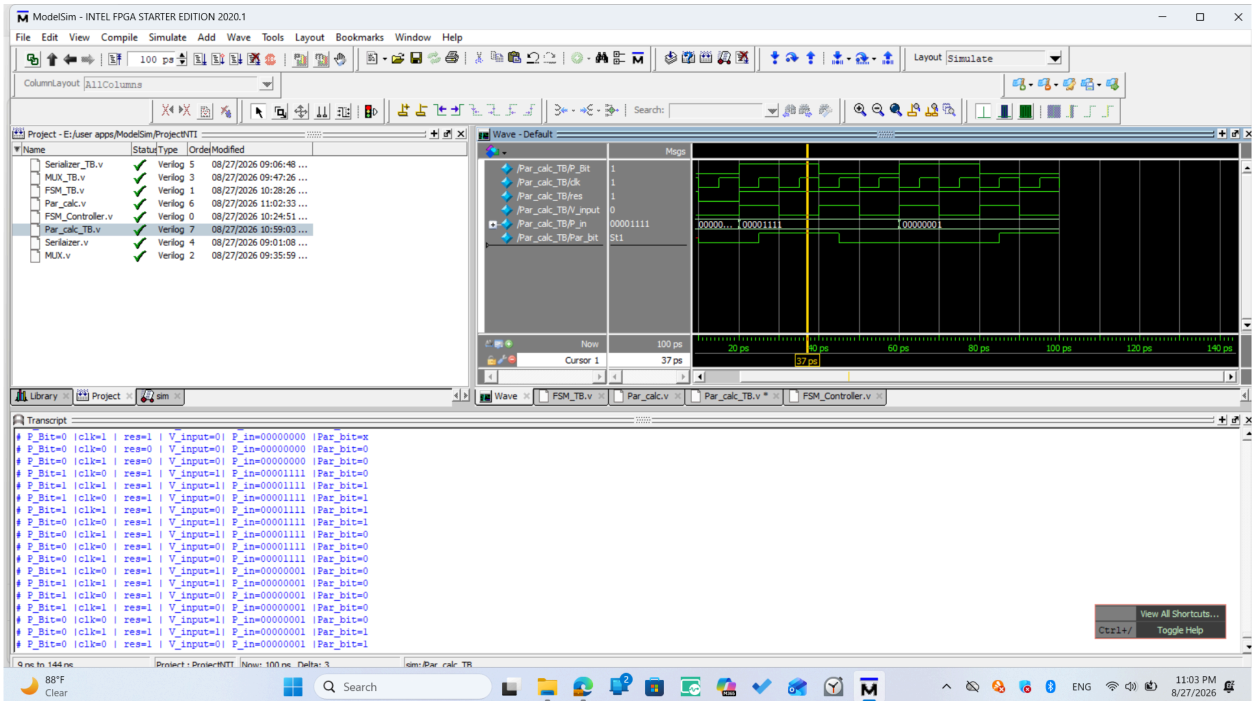


2.4 Parity Calculator Sub-Module Verification

Figure 5: Parity Calculator Waveform

The Parity Calculator derives parity status bit Par_bit based on mode flag P_Bit.

- **Execution Flow:** Mode P_Bit = 0 calculates Even Parity, while P_Bit = 1 calculates Odd Parity over P_IN.
- **Waveform Commentary:** For P_IN = 8'b00001111 (four set bits), Even Parity yields Par_bit = 0 and Odd Parity yields Par_bit = 1.



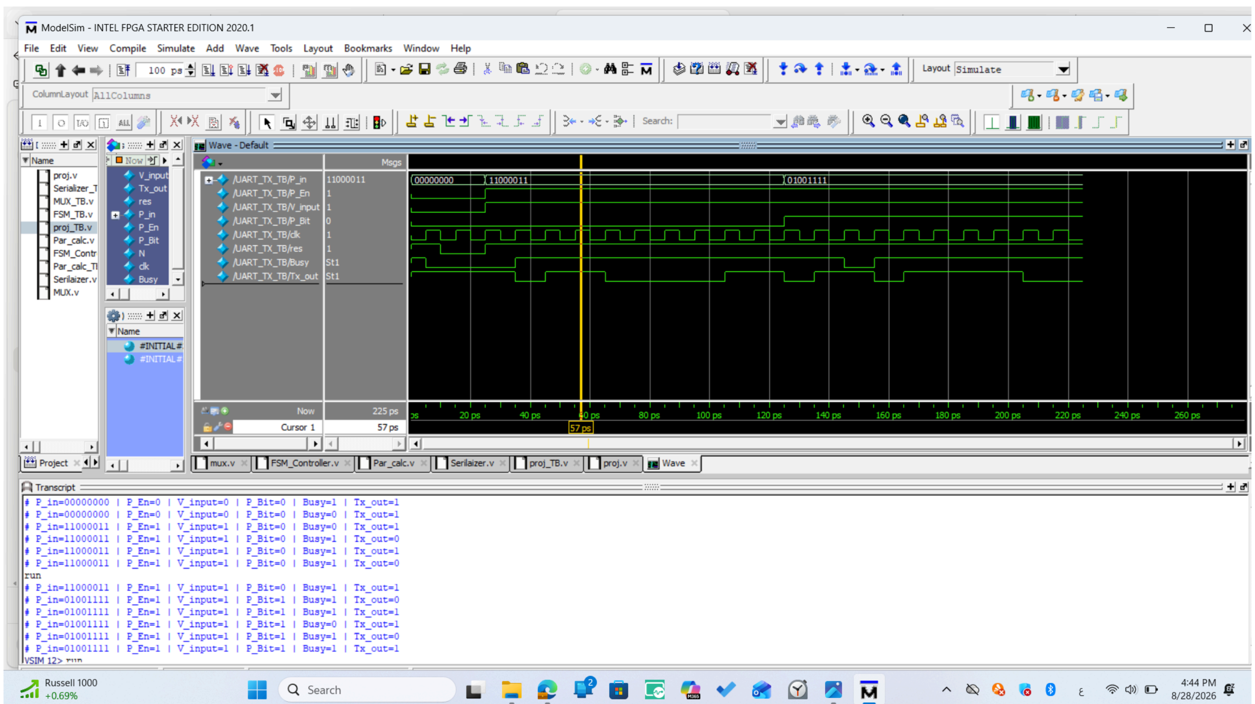
3. Top-Level Integration & Simulation Results

Figure 6: Integrated UART_TX Top Waveform

Full integration verification evaluates end-to-end frame serialization across complete transfer cycles.

Simulation Sequence:

1. **Idle State:** Inputs P_IN = 8'b11000011, P_Bit = 0. System holds TX_out = 1 (STOP line) and Busy = 0.
2. **Transmission Start:** Asserting V_input = 1 triggers Busy = 1. The multiplexer outputs Start bit (0), followed by LSB-first serial data bits (1, 1, 0, 0, 0, 0, 1, 1).
3. **Parity & Framing Completion:** The frame completes with parity bit insertion (0 for Even Parity) and Stop bit (1), after which Busy returns to 0.



4. Conclusion

The RTL implementation of the parameterized UART_TX module was synthesized and verified successfully. Unit testing confirmed functionality across all sub-modules (Serializer, MUX, FSM_control, and Par_calc), and top-level integration verified compliance with standard serial transmission framing protocols.